

# RQL Combustion using Ammonia \*

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## Abstract

This project examines the aeroelastic performance of a general aviation airplane wing for compliance with regulatory requirements, focusing on flutter analysis and design improvements. If needed, modifications are proposed and evaluated for structural, aerodynamic, and environmental impacts. Additionally, the feasibility of an all-electric version is analyzed, addressing design changes and economic and environmental implications. Model validation ensures the reliability of the tools.

*Keywords:* Aeroelasticity, Flutter Analysis, Dynamic Stability, Aircraft Wing Design

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## 1. Background

This section talks about the process of the RQL reaction mechanism.

Boundary conditions:

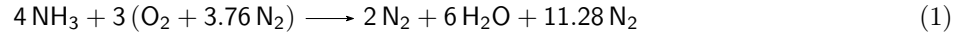
T in fuel = 500 K

T in air = 800 K

P in fuel = 15 bar

P in air = 12 bar

Writing the balanced chemical equation



In Equation 1, the oxidizer nitrogen is kept separate from the inert air nitrogen to keep track.

## 10 2. Combustor Design

for the secondary flow, optimum value for the momentum flux;

$$J_{opt} = 40$$

### 2.1. Swirler Design

Vane angle 30 - 60 deg

vane thickness 0.7-1.5 mm

number of vanes 8-16

## 3. Methodology

### 3.1. CEA/RPA or Cantera

San Diego model used with a nitrogen sub mechanism.

[1]

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\*Fully documented code is available on  [github/Combustion-W2025](https://github.com/Combustion-W2025). Short developer guide presented in the Appendix.

### *3.2. Numerical Simulation*

## **4. Results and discussion**

yahan kuch likna hai

### *4.1. Cantera*

Cantera results headers

### *4.2. Simulation*

Simulation results here

## **References**

- [1] University of California at San Diego, Mechanical and Aerospace Engineering, Chemical-kinetic mechanisms for combustion applications, <https://web.eng.ucsd.edu/mae/groups/combustion/mechanism.html>, accessed: 2023-10-01.

## Appendix A - Published results of p and p-k method

## Appendix B - Program listing for the main pk method logic